

SU(3) x SU(2) x U(1): The residual symmetry of extended conformal gravity

Source-backed arXiv sample

1 Extracted Lists

The list environments below are extracted from arXiv LaTeX source and wrapped for pdfLaTeX validation.

1. Conformal symmetry must underlie our mathematical description of nature.
2. Naive conformal gravity fails.
1. Scale changes, generated by the dilation operator, D directly produce the allowed choices of units by rescaling lengths locally.
2. Conformal translations. The four generators, K_a , translate infinitesimal vectors in inverse coordinates. Such a vector is inverted through a unit sphere, translated, then re-inverted. The transformation clearly involves an arbitrary displacement vector. The thing to notice – or rather not to notice – is that the vector is also rotated and rescaled. Since both rotating and stretching are already independent transformations, the group closes.